FOR THE STATE — THE SOVEREIGN BRIEF

THE ABSENCE AUDIT — THE METHOD, AT NATIONAL SCALE

Tunisia, audited — the state program, complete

The Fed transmission measured, the program architecture, the instruments, the financing, the gates, the signature path — every line cited.

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1. The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **$40.46B**, **78.7% of GDP**; debt service **26.2% of export revenues**; reserves **$9.34B** = 4.3 months of imports; food $3.00B + fuel $4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

Channel 1 — the debt trap: the “original sin” (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

Channel 2 — imported inflation: the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food ($3.00B) and fuel ($4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

The loop’s answer: owned export engines earn ~$117M/yr of gross exports outside the Fed cycle; the substitution legs take $152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 → BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

2. The state program — architecture, instruments, financing, adoption

Counterparts — who signs what

INSTITUTION	INSTRUMENT	ROLE
GCT (Tunisian Chemical Group)	acid-stream MOU + sampling agreement	the treasury's feedstock
SONEDE	municipal/industrial water offtake at >=$0.68/m3 gate	water contracts under the existing regime
STEG	25-year PPA under the IPP regime (500 MW class)	the power leg's revenue
Ministry of Industry, Mines & Energy	program sponsor; tender registration	the executive mandate
Ministry of Health	biomonitoring + cohort registry	the scoreboard

INSTITUTION	INSTRUMENT	ROLE
CNSTN	NORM licensing for the uranium route	the regulatory pass
Ministry of State Domains & Agriculture	40-year leases under Law 93-120 (86,000 ha available)	the land
BCT	reserve/rating path; FX policy coordination	the monetary endgame

Legal instruments — nothing new needs to be invented

INSTRUMENT	WHAT IT PROVIDES	CITATION
Organic Law 2016-35 (BCT)	monetary autonomy; the program does NOT monetize	Pg4/Pg5
IPP regime — Law 2015-62 & 2024/25 awards	renewables concessions; ~$31/MWh PPAs; 500 MW precedent	Rc13
PPP framework — Law 2015-49	desal/metals as PPPs where the state is the offtaker	Pg2
Law 93-120 (state-domain leases)	40-year agricultural leases; ~86,000 ha earmarked	Rc19
NORM/radiological regime (CNSTN, IAEA standards)	the uranium route's license layer	Pg11/Rm12

Financing stack — staged, gated, modeled in code

STAGE	FINANCING	GATE / CONDITION
Phase 0 ($150-253k)	grant / founder / early mandate	no DFI needed; 90 days
Health pilot ($58k)	ESG/grant	the first scoreboard numbers
Metals plant (~$100M)	60% debt @7% + offtake security; MIGA	DSCR >=1.2 at contract prices (nations_v5)
Solar (500 MW, ~$550M)	DFI stack: EIB/AfDB/WB-IFC at 4.5% + MIGA; or lean build	DSCR gate inside the bankable region (nations_v5)
Desal (~$165M)	70% debt @6% + municipal offtake + sovereign backstop	blended >=$0.68/m3, >=60% pre-sold
Terfas + food ($7-35M)	concessional agri + founder	nursery first
EU Global Gateway (17 GW)	the national envelope this program plugs into	Pg7

Governance

- every gate is a public number — published, dated, re-checkable
- health fund separate from the industrial SPVs; one shared data platform
- quarterly scoreboard: water chemistry, filter uptime, cohort KPIs, price index in Gafsa/Gabes
- each pilot passes or dies alone, in public — the falsification ladder is the governance
- commercially-validated ledger published at zero until the first signature moves it

Adoption path — what the state signs, in order

1. **Month 0:** the state signs: GCT sampling MOU + Phase 0 mandate ($150-250k, 90 days, no construction)
2. **Day 90:** two anchor measurements land: the 100-point water panel and the acid-stream assays — every projection becomes a signed number
3. **Months 4-18:** pilots on public gates: acid pilot, 100 ha terfas + nursery, spirulina pond, solar bid, desal offtake, health KPIs
4. **Years 2-5:** build only what passed: metals, solar, desal, health at basin scale — module by module, each with its own financing
5. **Years 5-10+:** the full loop: municipal spine, compute colocation, hydrogen, brine valorization — the destination the original design described

3. The household ledger — what this does to the cost of living

A state program that does not make life cheaper for its people is decoration. Each line, computed from the same primitives as the rest of this document; the quarterly price index in Gafsa and Gabès is the published enforcement: **the price of protein and vegetables must bend downward within 24 months of the first harvest.**

HOUSEHOLD LINE	TODAY	AFTER THE LOOP
Water	$1.00-1.50/m³ (SONEDE produced)	$0.35/m³ coastal · $0.34/m³ inland
Protein	$15-25/kg imported	$5-10/kg local
Terfas	export €20-40/kg	one third stays home at ≤€12/kg, contractual
Health	the disease's bill	$2,592 per healthy year vs the $4,000/DALY standard
Energy (grid)	$0.110/kWh industrial	$0.031/kWh for the loop's loads; tariffs stay state policy
Work	28% unemployment (54% women graduates)	workshop, nursery, ponds, plants, crews

3. The energy ledger — what power costs, per party

Anchors cross-checked: Rw10 (0.352 TND/kWh industrial, 19 TWh), Pg14 (the government's own 1,100 GWh = 5% of production, ~$125M/yr gas saving), Rc13 ($31/MWh inside the $24-35 historical bid band), Pg15 (fuel imports $4.895B, WITS-verified).

PARTY	TODAY	AFTER THE LOOP	THE DELTA
The loop's industries	grid $0.110/kWh	**$0.031/kWh** on the loop's PPA	−72%
The state	19-22 TWh/yr, ~95% gas, $4.90B fuel bill	876 GWh/yr solar (4.0-4.6% of production)	−$106-125M/yr gas imports ($114-142/MWh band)
The household	subsidized STEG tariff	the cost stack beneath the tariff falls	the inflation channel throttled — every dinar protected

3. The frontier — where the loop reaches, and where it honestly stops

The complete household budget, swept — nothing hidden, nothing promised beyond the frontier.

HOUSEHOLD LINE	STATUS	THE HONEST NOTE
Electricity (household tariff)	**indirect**	household tariffs stay subsidized STEG policy; the loop lowers the cost stack they sit on and the subsidy burden the state carries
Cooking gas (the bonbonne)	**out of scope**	the loop displaces grid gas, not bottled LPG; the bonbonne subsidy is a separate state decision
Transport fuel	**out of scope today**	the hydrogen leg serves ammonia for the fertilizer chain; transport fuels are a later policy question
Wheat and bread subsidy	**out of scope**	the protein stack complements the grain program; it does not replace it
Water	**covered**	cost floor $1.00-1.50 -> $0.35 coastal / $0.34 inland
Health	**covered**	$2,592 per healthy year vs the $4,000 standard
Protein	**covered**	$5-10/kg local vs $15-25/kg imported
Work	**covered**	the workshop, nursery, ponds, plants, crews where unemployment is 28% (54% for women graduates)
Housing	**out of scope**	the loop does not build housing; named so it is not promised

The water stack — line-by-line, with each credit's failure mode

The $0.347/m³ audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the $0.68 gate.

LINE	$/M³	PROVENANCE	WHAT MAKES IT FAIL
Base — solar energy 4 kWh/m³ at the loop's own PPA + capex + O&M	$0.567	MEASURED anchors (Rc13/Rw2/Rw3)	holds — if even the PPA lapses, grid power takes the base to $0.88, still under SONEDE
DC reject heat → RO feed preheat — one seawater pipe cools the servers, then arrives warm at the membranes (SEC −2.5%/°C)	−$0.007	DERIVED — Leg 03 + Leg 01, same pipework	fails only if no compute anchor tenant — and costs $0.007
Kiln/exotherm preheat — the acid plant and board kilns warm the same feed	−$0.000	DERIVED — Leg 04/05 waste heat	negligible by design — carried at zero margin
Brine valorization — salt/Mg evaporated on the loop's own free heat, sold at the park gate (of-take signed; nothing hauled)	−$0.137	DERIVED — brine is the desal's own by-product	if the of-take is not signed by FID, the credit degrades to −$0.03 (disposal avoided only) — the biggest single sensitivity
Shared civil works — one intake, one outfall, one grid connection serves RO + DC + metals in the same park	−$0.032	DERIVED — 12% capex sharing	if the park phases one module at a time, halve it to −$0.016

LINE	$/M³	PROVENANCE	WHAT MAKES IT FAIL
Dispatch value — the desal's own flexibility follows the loop's own solar curve	−$0.015	DERIVED — 20 GWh shifted	fails to zero if the STEG contract does not reward flexibility — contractual, not hoped
ESG financing delta — the health leg's public KPIs reprice the loop's own debt (100–150bp)	−$0.026	DERIVED — nations_v3	fails to zero if lenders do not accept the KPI track record — bankable-region fallback keeps the module alive
Brine pump-as-turbine — the desal's own residual pressure	−$0.002	DERIVED — 2.5 bar × 75%	fails to zero — immaterial
Integrated	**$0.347/m³**	**28% of SONEDE's $1.00-1.50 (Pg16)**	**the degraded cascade: if every soft credit fails, water still lands at $0.521/m³ — under SONEDE's own $1.00–1.50, and the $0.68 gate still clears with margin. The credits are the margin, not the plan.**

The AI spine — the system's nervous system, and its boundary

The AI runs the measurement, the routing and the publication — never the decisions.

COMPONENT	WHAT IT DOES	WHAT IT IS WORTH
The dispatcher	routes every joule and m³ in real time: solar → desal → electrolyzer → ponds → DC cooling, following the solar curve minute by minute	worth the 20 GWh/yr of dispatch value — the AI is what collects it
The scoreboard	computes and publishes every gate number automatically — filter uptime, urinary F, DSCR, yield records — tamper-evident, no human can edit a gate	the public contract, machine-audited
The price index	tracks the Gafsa/Gabès household basket quarterly and publishes the bend — the people's enforcement metric, computed not curated	the number that proves the prices bend, or proves they do not
The cohort engine	runs the 3,000-family health baseline, dose-outcome analytics and anomaly flags for the clinical team	the DALY ledger becomes a live dataset
The dispatcher twin	a digital twin of the loop (energy, water, brine, heat) that simulates before any physical change and audits after	the falsification ladder, automated
The boundary	the AI never sets a gate, never signs a contract, never prices a tariff — it measures, routes and publishes. The decisions stay human, in public.	the guardrail that keeps the nervous system a servant, not a master

The results framework — what a World Bank board reads first

The scoreboard IS the M&E system (PforR class, Pg23).

INDICATOR	BASELINE	TARGET	DATA SOURCE
Filter uptime, pilot households	0% (filters not installed)	≥70%	scoreboard, live
Urinary fluoride, pilot children	baseline to be measured (Day 1-30)	−20%	cohort engine

INDICATOR	BASELINE	TARGET	DATA SOURCE
Household price index, Gafsa/Gabès	baseline Q0	bend downward ≤24 months after first harvest	price index, quarterly
Water produced cost	SONEDE $1.00-1.50/$m^3$ (Pg16)	≤$0.35/$m^3$ integrated	plant telemetry
Energy recovered inside the loop	0 GWh/yr	33.9 GWh/yr + 20 GWh dispatch	dispatcher ledger
Gas imports displaced	$4.895B fuel bill (Pg15)	−$106-125M/yr	WITS Comtrade, annual
Uranium recovered from the acid stream	0 (lost to the sea)	≥85% recovery gate	assay chain
Terfas hectares planted	0 ha	100 ha gate → 2,500-3,000 ha	nursery registry
Jobs in the basin	28% unemployment (54% women graduates)	workshop → nursery → plants → crews	employment registry

The people's energy is electrons, not hydrogen — and the H&sb2; lives inside the loop

The loop's H&sb2; (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H&sb2; at $0.21 is 7× the loop's $0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H&sb2; volume equals 0.0% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m^3/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

The total routing — every stream gets a home

The ponds do *not* drink brine: SWRO brine is 55-70 g/L, past Arthrospira's ~30 g/L ceiling (Pg31); they drink the desal permeate and breathe the electrolyzer's oxygen. The full inventory:

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
Solar electrons	500 MW fleet	desal, metals, DCs, electrolyzer, pumps — routed by the dispatcher	876 GWh/yr
SWRO permeate	coastal desal	city offtake · electrolyzer feed · ponds · greenhouses	150,000 m^3/d
SWRO brine	coastal desal	salt + magnesium valorization, evaporated on the loop's own free heat	122,727 m^3/d → 58 kt Mg/yr
Brine bittern	valorization	Mg(OH)&sb2; fire retardant → **the board's own chemistry**	140 kt/yr = 140% of the board's need

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
Electrolyzer oxygen	H&sb2; plant	spirulina pond + greenhouse aeration	21 kt/yr
Electrolyzer hydrogen	H&sb2; plant	ammonia → the GCT fertilizer chain	2.63 kt/yr
Electrolyzer reject water / heat / flush	H&sb2; plant	product line / desal preheat / brine line	nothing leaves
DC reject heat	compute park	the same seawater pipe, arriving warm at the membranes	−$0.007/$m^3$
Kiln + acid exotherm	board kilns + metals plant	desal preheat + greenhouses	30 GWh/yr
CO&sb2; (kiln + power)	combustion streams	mineralized into board (2 Mt PG) · pond carbon	540 t CO&sb2;/yr to the ponds
Phosphogypsum	GCT stack	fire-rated board — with its retardant from our own brine	2 Mt/yr
Acid stream	GCT wet-process acid	U + heavy REE extraction; raffinate back to fertilizer	716.5k lbs U&sb3;O&sb8;/yr
Municipal return flows	city mesh	30%+ reuse · pump-as-turbines on the descent	3.9 GWh/yr
Sludge / digestate	wastewater + organics	biogas for kilns and greenhouses	booked option, honestly labeled
Slaughterhouse bones	local slaughterhouses	bone char → the fluoride filters; **spent char → phosphate soil amendment for the fields**	filter loop closes into fertilizer
Spent pond medium	spirulina harvest	nutrient-rich → greenhouse fertigation	nothing discarded
Ammonia fertilizer	GCT chain	the terfas fields' host plants + greenhouse soils	the truffle grows on the loop's own nitrogen
Compute services	colo park	export — the data product	tenant-gated
Ra-226 / Po-210	acid stream radiology	stays in the gypsum/board matrix — the radiology gate	below the 1,000 Bq/kg exemption
Board offcuts	board line	recycled into new board	zero waste
Sodium chloride	brine line	industrial offtake at the park gate	nothing hauled
Halophytes / Artemia on brine	brine line	future option, honestly labeled — not a current credit	the only stream awaiting its consumer

the ponds drink the desal permeate — ~{:.0f} m^3/d, {:.2f}% of the 150,000 m^3/d output — and breathe the electrolyzer's oxygen; the brine (55–70 g/L, far past the ~30 g/L ceiling the organism tolerates) is routed to salt and magnesium valorization instead. Halophytes and Artemia on brine are booked as a future option, honestly labeled — not a current credit.

The fuel invoice, honestly split — the 500 MW is the grid's medicine, not the whole prescription

of the $4.895B fuel invoice, the 500 MW fleet displaces only the power-sector gas slice — ~$0.75B/yr (DERIVED from the energy balance) — by $106-125M/yr, which is 14133-16667% of that slice. The rest of the invoice has its own answers, not solar's: the bottles (~$0.55B of LPG, the bonbonne) leave the market as the people shift to electrons — induction on the loop's own surplus — and the biogas pilot; transport fuels (~$2.6B) are the honest frontier. The 500 MW is the grid's medicine, not the country's entire prescription.

The mountains, honestly analyzed

Fog collection is real where fog lives (10-20 L/m²/day, Pg33) — Tunisia's fog lives in the already-humid NW; the truffle mountains are the arid interior, and cloud seeding has no proven routine gain (Pg32). What the mountains give the loop: the mountains enter the loop as storage, rainfall and free cooling — and as the land where the truffle belongs (freeing the lowlands that grow today's food). Fog harvesting from pipe meshes stays off the program: the moisture is not there to collect, and the honest scientist does not plant a net under a cloudless sky.

4. What the state receives

External account: ~$279M/yr of exports + import substitution + financing delta at full phase — measured against the 26.2% of export revenues that debt service takes today. **Rating path:** B-/Caa1 → BB- and reserves 4.5 → 6+ months as the owned-export engines compound (Rc11). **Health scoreboard:** the basin's water chemistry, DALY ledger and filter uptime become public KPIs — the collateral that lowers the price of every loan. **Jobs:** workshop, nursery, ponds, plants, solar crews, health workers — in the towns where the 2008 and 2015 blockades began. A job is the cheapest peace treaty ever signed.

5. The gates are public — the state sees every number

MODULE	GATE	PASSES ON
Health pilot (100 households, 2 schools, 90 days)	filter uptime ≥70%, urinary F ↓20%	public KPIs
Metals pilot	U recovery ≥85%, all-in ≤$75/lb, offtake signed	GCT stream
Solar	DSCR ≥1.15 inside the bankable region, or grant support	PPA award
Desalination	≥60% contracted offtake, blended ≥$0.68/m³	brine permit
Terfas	3-season record ≥200 kg/ha at ≥€20/kg; quota clause signed	scaling toward 2,000-3,000 ha

6. Phase 0 — what the 90 days produce for the state

Two anchor measurements — the 100-point water panel and the acid-stream assays ($75k of the $253k list) — plus the emissions inventory. Every projection on this page becomes a signed number before one brick is laid. Nothing is asked of the state beyond cooperation, and every gate is a number the state can check.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

Works cited

1. [Rm17] GCT throughput (~6.5 Mt phosphate rock/yr; >10 Mt/yr PG at peak; 'hundreds of millions of tons' discharged at Gabès) — http://www.gct.com.tn/fileadmin/user_upload/Downloads/Rapport_Annuels/GCT-ANNUAL-REPORT-2012-PART1.pdf...
2. [Rw7] Tunisian water tariffs: SONEDE 0.200-2.310 TND/m³; irrigation 0.060-0.160 TND/m³; groundwater 0.112-0.162 TND/m³ — https://www.sonede.com.tn/tout-savoir-sur-leau/la-facture-deau-methodes-de-paiement/...
3. [Rw10] Tunisia grid: STEG 5.9 GW, ~19 TWh/yr, 95% gas-fired, renewables <5%, industrial tariff 0.352 TND/kWh; budget deficit 5-7% GDP, debt ~83% GDP — https://www.trade.gov/country-commercial-guides/tunisia-electrical-power-systems-and-renewable-energy...
4. [Rw12] Tunisia land law: foreign ownership of agricultural land prohibited; 40-year leases only; state-domain conversion is an administrative hurdle — https://2009-2017.state.gov/e/eb/ifd/2008/101021.htm...
5. [Rc11] Tunisia sovereign ratings: Moody's Caa1 (Feb 2025 upgrade, stable); Fitch B- (Sep 2025, affirmed Jan 2026); reserves ~4.5 months import cover; WACC 10-15% — https://www.fitchratings.com/research/sovereigns/fitch-upgrades-tunisia-to-b-outlook-stable-12-09-2025...
6. [Rc12] Who finances Tunisia: EIB, AfDB, KfW, World Bank/IFC, EU Global Gateway (17 GW renewable programme); no active IMF program — https://international-partnerships.ec.europa.eu/policies/global-gateway/17-gw-renewable-energy-programme-tunis...
7. [Rc13] Empirical IPP record: 500 MW solar awarded 2024/25 (Scatec, Qair, Voltalia 25-yr PPAs at ~$0.031/kWh); installed renewables ~400 MW; STEG debt >$1.3B — https://www.engineeringnews.co.za/article/scatec-awarded-ppas-for-wind-solar-projects-in-tunisia-2026-01-26...
8. [Rc19] Tunisia arid land: ~86,000 ha of state domain earmarked for 40-year agricultural leases (Law 93-120) — https://documents1.worldbank.org/curated/en/926441468778187741/pdf/multi0page.pdf...
9. [Ru2] URANEXT/OCP (Morocco) building the first modern yellowcake-from-phosphoric-acid plant, ~$100M CAPEX, El Jadida, 2025 — https://northafricapost.com/87963-uranext-to-build-moroccos-first-yellowcake-uranium-plant-in-el-jadida.html...